BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

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Abstract

Recovering prior client-specific advice from a growing legal consultation archive is difficult under same-domain interference. Dense retrieval often ranks the client's question highly while missing the lawyer's answer—an episode-incompleteness failure—and product-quantized indexes can further distort turn scores at moderate archive size, fragmenting consultation episodes across unrelated top-k slots in a vector database. BIMS-LEGAL is a dual-store Brain-Inspired Memory System for Chinese legal consultation logs. Its systemic contribution is a complementary *episodic store* (timestamped turns with role and session identifier *sid*) paired with a slower *BIRCH semantic store* of thematic clusters, separating fast episode recovery from cross-session semantic integration. Algorithmically, Soft O2 (session soft binding) and Soft O2-C with gated Hybrid cluster binding address episode incompleteness under IVFPQ score collapse and context fragmentation: siblings inherit attenuated scores ($\beta \cdot s$) rather than hard score copy, completing question-hit / answer-miss episodes while preserving rank competition. We formalize Soft O2 as a three-candidate ranking-competition operator and derive a corpus-adaptive $\beta^\star$ from pairwise hinge and soft-rank losses; fitted gap ratios on LegalEp stores place $\beta^\star$ in $\{0.95, 0.98\}$, matching the empirical Answer-Hit peak band. O1 (exact FlatIP indexing) and O3 (deferred bulk-load consoli-

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dation) are implementation and scale design operators that enable rigorous large-scale shared-store experiments; they are not standalone algorithmic contributions. We evaluate on CAIL2024 multi-turn dialogues and on LegalEp episode corpora rebuilt from DISC-Law-SFT and Lawyer-LLaMA, with exact replay as a diagnostic channel and paraphrase, follow-up, and advice-recall as primary channels. Controls include FlatIP, hard parent hydration, BM25, dense–sparse RRF, and cross-encoder reranking, with McNemar tests and bootstrap intervals. On the M=400 shared store, O1+Soft O2 raises Answer Hit from 0.540/0.397 to 0.780/0.680 on DISC/Lawyer. On CAIL multi-turn Soft O2 lifts Answer Hit by +0.22 to +0.52 over FlatIP with collapsing shuffled-*sid* controls. On same-session LegalMem/LegalEp stores Soft O2 dominates ungated Soft O2-C; under a fair Mix reconstruction of the same stores, gated Hybrid Soft O2+Soft O2-C improves over Soft O2 on LegalEp-Lawyer (AH 0.534 vs 0.496, mid-p=0.010).

Keywords: legal information retrieval, consultation memory, soft episode binding, dual-store memory, cluster binding, Chinese legal corpora

1. Introduction

A junior associate may ask a firm assistant: "Last month you advised this client on terminating a fixed-term contract—what remedy did we recommend?" The system returns several near-identical "contract termination" questions, ranks the stored user utterance highly, and never surfaces the partner's written advice. The subsequent reply may sound fluent yet be grounded in the wrong episode—or in none. In consultation archives, questions and answers share specialized terminology; neighbouring matters are topically adjacent yet legally distinct; and the professionally useful evidence unit is the full consultation *episode*.

Legal informatics has long studied statutory interpretation, case retrieval, and advisory systems Martinez-Gil (2023); Ashley (2017). Recent legal large language models (LLMs) improve statutory and fact-pattern answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical legal information retrieval (IR) mainly targets *external* authority—statutes, cases, or community answers—as documents Askari et al. (2024); Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented dialogue systems recover evidence from evolving conversation logs Wu et al. (2024); Maharana et al. (2024); Packer et al. (2023). The gap addressed here is *internal* *consultation memory*: retrieving prior client-specific advice from a growing same-domain archive under lexical interference.

BIMS-LEGAL is a dual-store Brain-Inspired Memory System for legal consultation archives, designed and evaluated in this study. The *systemic contribution* is a dual-store architecture that separates a fast episodic pathway (timestamped turn embeddings with speaker role and session identifier *sid*) from a slow semantic pathway based on Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH) with deferred consolidation Zhang et al. (1996). Complementary Learning Systems (CLS) theory McClelland et al. (1995); Kumaran et al. (2016) supplies an architectural analogy for this dual-pathway design, not an empirical claim about biological memory. Under same-domain interference, dense turn retrieval in a vector database often returns a question turn while its answer sibling remains outside top-k—a system-level episodic memory failure we term *episode incompleteness*, distinct from a full session miss and exacerbated by IVFPQ score collapse and context fragmentation at moderate archive sizes.

The *algorithmic contributions* are soft binding operators that complete episodes without hard score copy:

- **Soft O2 (session soft binding).** When a turn is retrieved with score s, siblings sharing *sid* inherit $\max(s_{t'}, \beta \cdot s)$, surfacing answer turns under question-hit / answer-miss while preserving rank competition among candidates.

- **Soft O2-C and gated Hybrid (cluster soft binding).** The same soft-inheritance rule applies within BIRCH clusters when gold evidence is not co-session with the query; Hybrid triggers cluster binding only on direct dense hits to avoid displacing same-session siblings.

Two **implementation and scale design operators** support rigorous large-scale shared-store experiments but are not standalone algorithmic contributions:

- **O1 (exact FlatIP indexing).** At hundreds-to-thousands of turns, IVFPQ product quantization is under-trained and collapses Answer Hit; FlatIP restores faithful turn scores so Soft O2 inherits uncorrupted sibling signals.

- **O3 (bulk-load consolidation).** Deferred BIRCH and disk writes make archive construction tractable without changing retrieval semantics.

We ask four research questions:

- **RQ1.** Under same-domain interference, does episode incompleteness (question hit / answer miss) dominate full session miss?
- **RQ2.** Do Soft O2 and Soft O2-C address episode incompleteness under measured index distortion, and what incremental gains does Soft O2 provide over FlatIP and hard parent hydration?
- **RQ3.** On the *same* LegalEp /LegalMem stores, when is the slow (BIRCH) pathway effective? Does Soft O2-C help under standard same-session gold, and under a fair Mix of same-/cross-session gold?
- **RQ4.** Across CAIL multi-turn and LegalEp paraphrase /advice-recall channels, how large and session-tied are Soft O2 gains, and how do Soft O2 and Soft O2-C /Hybrid complement each other?

Contributions..

1. A dual-store memory architecture (episodic + BIRCH semantic) for Chinese legal consultation archives, separating fast episode recovery from cross-session semantic integration under same-domain interference.
2. Soft O2 (session soft binding) as a system-level solution to episodic memory incompleteness under IVFPQ collapse and context fragmentation in vector databases, with Soft O2-C and gated Hybrid cluster binding for cross-session episode recovery.
3. A ranking-competition derivation of the Soft O2 attenuation β from pairwise hinge and soft-rank objectives on a three-candidate (hit /sibling /distractor) model, with gap-ratio fits that recover the empirical $\beta \in \{0.9, 0.95, 0.98\}$ operating band.
4. Same-store and fair Mix validation showing when each pathway applies: Soft O2 dominates when gold shares *sid*; gated Hybrid Soft O2+Soft O2-C improves under cross-session Mix reconstructions (LegalEp-Lawyer AH 0.534 vs 0.496, mid-p=0.010).
5. Multi-channel evidence on CAIL and LegalEp with BM25, RRF, and cross-encoder controls, McNemar tests, scale curves to M=6400, and a dual-protocol end-to-end QA audit linking complete episode retrieval to reduced hallucination.

6. O1 and O3 as implementation and scale design operators enabling the shared-store ablations (M=400, S=300) and archive construction at scale, reported separately from the algorithmic claims.

Section 2 reviews related work. Section 3 presents BIMS-LEGAL with formal dual-store operators, Soft O2 /Soft O2-C /Hybrid algorithms, and the $\beta^\star$ derivation. Section 4 details corpora and protocols. Section 5 reports results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.1. Legal IR and long-horizon dialogue memory

Legal QA and retrieval systems typically ground generation in statutes, cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware similar-case ranking further show how interaction context and structured legal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024). Those pipelines optimize access to external authority. Consultation memory retrieval targets a different evidence source: *prior consultation episodes* inside an agent's own archive. Confusing two clients' histories can produce fluent yet professionally inappropriate responses even when statute retrieval is correct. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM improve statutory and advisory generation Yue et al. (2023); Huang et al. (2023); Yue et al. (2024), but they still need a reliable memory index when prior advice must be recovered from a growing same-domain log.

Legal consultation archives add a further difficulty: high lexical overlap among topically adjacent matters, with the professionally useful evidence unit being the full question–answer episode. This setting motivates *dual-store* memory architectures that keep fast episodic turn retrieval separate from slower semantic clustering, and session-aware soft binding operators that recover answer turns under same-domain interference without collapsing rank competition among unrelated candidates.

2.2. Dual-store memory and episodic retrieval

Dual-store designs distinguish fast episodic encoding from slower semantic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving et al. (1972); Atkinson and Shiffrin (1971). Memory architectures for dialogue agents extend context via hierarchical paging, summary banks, graphs, and neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Rasmussen et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang et al. (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved units as flat passages rather than *multi-turn consultation episodes* whose answer turn may rank below an unrelated question turn sharing legal vocabulary. LongMemEval and LoCoMo evaluate long-horizon recall and QA over personal conversation logs Wu et al. (2024); Maharana et al. (2024), while broader memory-agent surveys emphasize persistence beyond the context window Wu et al. (2025); Tan et al. (2025). Recent conversational RAG work stresses reuse of historical search evidence across turns Mo et al. (2026). BIMS-LEGAL positions its dual-store contribution at this intersection: episodic turn embeddings with *sid* for same-session recovery (Soft O2), and BIRCH centroids with Soft O2-C for cross-session recovery when query and gold evidence occupy different sessions but share thematic structure Zhang et al. (1996). Soft O2 directly addresses the episodic retrieval challenge that parent-document hydration and session-max aggregation handle via hard score copy: under dense same-domain interference, a question turn can dominate top-k while its answer sibling remains absent, producing fluent yet ungrounded downstream generation unless siblings enter under attenuated inherited scores.

2.3. Session-aware retrieval

Parent-document hydration, session aggregation, and joint session indexing are standard session-aware mechanisms for attaching sibling context to a dense hit. Soft episode binding (Soft O2) inherits $\beta \cdot s$ for siblings and keeps ranking competition among candidates. Lexical BM25 Robertson and Zaragoza (2009), dense passage retrieval Karpukhin et al. (2020), dense–sparse Reciprocal Rank Fusion Cormack et al. (2009), and cross-encoder reranking Nogueira et al. (2019); Chen et al. (2024) provide complementary controls outside session membership. Exact flat indexes and product-quantized ANN backends Johnson et al. (2019) further affect whether sibling binding is applied over faithful turn scores. Table 1 summarizes the session-structure contrasts used in the main grids.

[*] On the reported corpora, session-max rankings coincide with hard parent hydration under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2 denotes session soft binding; Soft O2-C denotes cluster soft binding on the BIRCH pathway.

Table 1: Session-aware mechanisms compared in this work.

Mechanism	**Unit**	**Expansion**	**Type**	**Rank**
Parent hydration	Session	Insert siblings	Hard	No
Session-max	Session	Session max	Hard	Same*
Joint episode index	Session doc	Atomic	N/A	N/A
Soft O2	Turn+*sid*	Soft βs	Soft	Yes
Soft O2-C	Turn+cluster	Soft $\beta_c s$	Soft	Yes

We reuse CLS vocabulary as an architectural analogy for BIMS-LEGAL: turn embeddings with timestamps, roles, and session identifiers as episodic elements; Soft O2 as episode binding on the episodic pathway; BIRCH centroids with Soft O2-C as semantic integration on the slow pathway; and shared-store distractors as interference. The dual-store split is the systemic design; Soft O2 and Soft O2-C are the algorithmic mechanisms evaluated on Chinese legal consultation archives.

3. Method: BIMS-LEGAL

3.1. Architecture overview

Figure 1 summarizes the pipeline. BIMS-LEGAL maintains complementary episodic and semantic stores over the same consultation archive.

Definition 1 (Episodic and semantic elements)**.** *An episodic element is* $e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, sid_i)$*, where* $\mathbf{v}_i \in \mathbb{R}^d$ *is an embedding,* t_i *a timestamp,* c_i *conversational context (including speaker role),* ϕ_i *surface features, and* sid_i *the consultation session identifier. A semantic element is* $s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$ *with centroid* $\mathbf{c}_j$*, member set* C_j*, label* κ_j*, and formation statistics* τ_j*. Write* $\mathrm{Sib}(t) = \{t' : sid_{t'} = sid_t\} \setminus \{t\}$ *for session siblings and* $\mathrm{Clus}(t) = \{t' : cid_{t'} = cid_t\} \setminus \{t\}$ *for BIRCH-cluster siblings.*

Definition 2 (Dual-store retrieval map)**.** *Given query* q *with embedding* $\mathbf{v}_q$*, base episodic retrieval returns a scored candidate set* $\mathcal{R}_0(q) = \{(t, s_t)\}$ *under Eq.* (1)*. Soft O2 produces* $\mathcal{R}_{\mathrm{O2}}(q) = \mathsf{SoftBind}(\mathcal{R}_0(q); \mathrm{Sib}, \beta)$ *(Section 3.3). Soft O2-C produces* $\mathcal{R}_{\mathrm{O2C}}(q) = \mathsf{SoftBind}(\mathcal{R}_0(q); \mathrm{Clus}, \beta_c)$ *(Section 3.5). Gated Hybrid composes session binding then cross-session cluster binding (Algorithm 2). The returned list is the top-*k *of the bound set under the rewritten scores.*

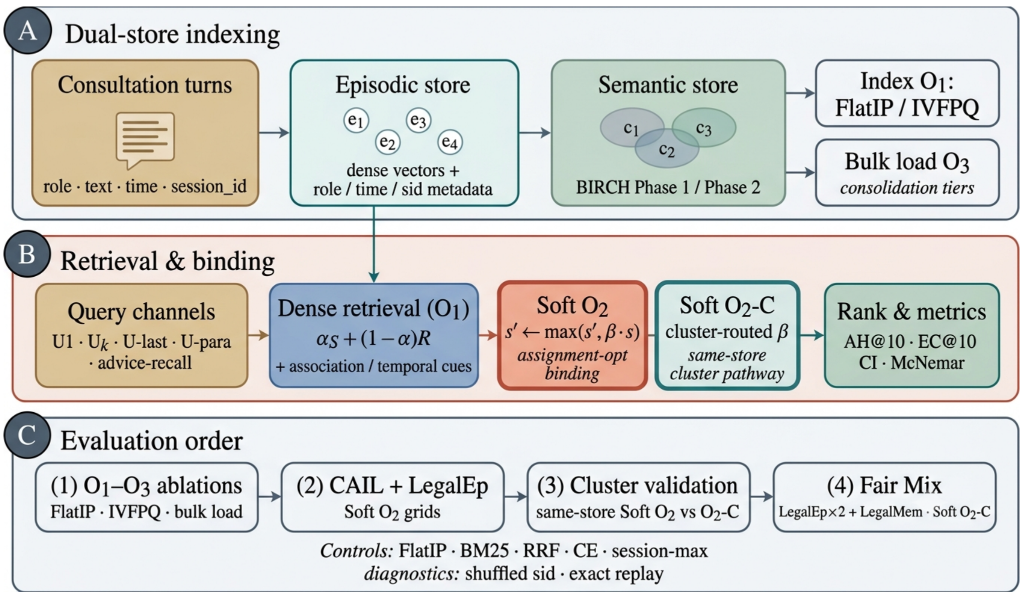

Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and BIRCH semantic clustering (O3 bulk-load consolidation). (B) Query-time dense retrieval under O1 FlatIP with Soft O2 session binding and Soft O2-C cluster binding (algorithmic core); O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, CAIL /LegalEp Soft O2 grids, same-store cluster validation, and fair Mix Soft O2-C.

3.2. Base scoring and implementation operators (O1, O3)

For query q with embedding $\mathbf{v}_q$, episodic base retrieval scores a memory turn m by

$$\text{score}(m) = \alpha\, S(\mathbf{v}_q, \mathbf{v}_m) + (1-\alpha)\, R(t_m), \tag{1}$$

where S is cosine similarity, $R(t) = \exp(-\lambda \Delta t)$ is an exponential recency prior, and $(\alpha, \lambda) = (0.7, \lambda_0)$ by default with α the semantic weight. Eq. (1) is the ranking score used throughout the legal grids; Soft O2 /Soft O2-C are subsequent binding operators and do not replace the base score.

O1: Exact FlatIP indexing.. Let $\widehat{S}$ denote the similarity induced by an ANN backend. At hundreds to a few thousand turns, FAISS IVFPQ is often under-trained, so $\|\widehat{S} - S\|$ is large enough to reorder turns and break Soft O2 inheritance. O1 forces $\widehat{S} = S$ via exact FlatIP (inner-product search on ℓ_2-normalized embeddings). Quantization error depresses Answer Hit (Table A.8: IVFPQ AH 0.330 at M=1600 on LegalEp-DISC exact). O1 is an engineering prerequisite for rigorous Soft O2 evaluation, not a standalone retrieval algorithm.

O3: Bulk-load consolidation.. During bulk construction, O3 replaces the per-insert update ($e_i \mapsto \mathsf{BIRCH}(e_i)$; Flush) by a deferred operator $\mathsf{Finalize} = \mathsf{BIRCH}(\{e_i\}_{i=1}^{N}) \circ \mathsf{Flush}$, executed once after all episodic writes. O3 does not change retrieval semantics; it makes $M \geq 400$ shared-store experiments runnable (≈ 11 min wall-clock for $M{=}400$; amortized ≈ 0.825 s/turn over ~ 800 turns). Quality–latency curves to $M{=}6400$ appear in Section 5.4.

3.3. Soft O2: session soft binding

Motivation.. Under same-domain interference, a retrieved question turn can score highly while its answer sibling remains outside top-k—episode incompleteness compounded by IVFPQ score collapse and context fragmentation. Soft O2 completes episodes without hard score copy.

Definition 3 (Soft binding operator)**.** *Given a scored set $\mathcal{R} = \{(t, s_t)\}$, a sibling relation $\mathcal{N}(\cdot)$, and attenuation $\beta \in (0,1]$, define*

$$\mathsf{SoftBind}(\mathcal{R}; \mathcal{N}, \beta): \quad s_{t'} \leftarrow \max\bigl(s_{t'},\, \beta \cdot s_t\bigr) \quad \textit{for all } t \in \mathcal{R},\ t' \in \mathcal{N}(t). \tag{2}$$

Soft O2 is $\mathsf{SoftBind}(\cdot; \mathrm{Sib}, \beta)$ *with default* $\beta{=}0.98$*. Hard parent hydration is the special case* $\beta{=}1$ *with forced insertion of all siblings; session-max replaces each member score by* $\max_{t \in sid} s_t$*.*

Algorithm 1 states the deployed Soft O2 procedure (expand + completeness truncate), matching the public implementation.

3.4. Ranking-competition derivation of β

We lift β from a heuristic near-unity constant to the optimum of an explicit ranking-competition loss on a minimal three-candidate model.

Definition 4 (Three-candidate episode model)**.** *For a query that densily retrieves a gold question turn h (the* hit*) with score $s_h > 0$, let a be the gold answer sibling and d the strongest same-domain distractor outside sid_h, with dense scores s_a and s_d. Soft O2 rewrites only the sibling:*

$$s'_h = s_h, \qquad s'_a = \max(s_a, \beta s_h), \qquad s'_d = s_d. \tag{3}$$

Algorithm 1 Soft O2 session soft binding

Require: Query q, base hits $\mathcal{R}_0$, top-k, attenuation β

$\mathcal{R} \leftarrow \mathcal{R}_0$
for each hit $t \in \mathcal{R}_0$ with score s_t **do**
  for each sibling $t' \in \mathrm{Sib}(t)$ absent from $\mathcal{R}$ **do**
    insert $(t', \beta \cdot s_t)$ into $\mathcal{R}$
  end for
end for
  ▷ Completeness pass after fusion with Eq. (1)
for each session *sid* represented in $\mathcal{R}$ **do**
  $s_{\max} \leftarrow \max\{s_u : u \in \mathcal{R},\ sid_u = sid\}$
  for each $t' \in$ Sib of *sid* still absent **do**
    insert $(t', \beta \cdot s_{\max})$ into $\mathcal{R}$
  end for
end for
return top-k of $\mathcal{R}$ by descending score

Define the relative interference (gap ratio)

$$\rho = \frac{s_d}{s_h} \in \mathbb{R}_+. \tag{4}$$

When s_a lies below the competitive fringe (the incompleteness regime), entry of a into top-k against d requires $\beta s_h > s_d$, i.e. $\beta > \rho$, while strict rank competition with the hit requires $\beta < 1$.

Pairwise hinge objective.. With availability margin $m_{\mathrm{a}} \geq 0$ and rank margin $m_{\mathrm{r}} \geq 0$, the per-query hinge loss is

$$\mathcal{L}_{\text{hinge}}(\beta; \rho) = w_{\mathrm{a}} \big[m_{\mathrm{a}} - (\beta - \rho) \big]_+ + w_{\mathrm{r}} \big[m_{\mathrm{r}} - (1 - \beta) \big]_+, \tag{5}$$

where $[\cdot]_+ = \max(\cdot, 0)$ and $(w_{\mathrm{a}}, w_{\mathrm{r}})$ trade Answer-Hit availability against hit-first rank preservation Burges et al. (2005); Joachims (2002). For $m_{\mathrm{a}} = m_{\mathrm{r}} = 0$ the population risk $\mathbb{E}_\rho[\mathcal{L}_{\text{hinge}}]$ is minimized by covering a π-fraction of interference cases,

$$\beta^\star_{\text{hinge}} = \mathrm{clip}(\mathrm{Quantile}_\pi(\rho),\ 0,\ 1)\,, \qquad \pi = \frac{w_{\mathrm{a}}}{w_{\mathrm{a}} + w_{\mathrm{r}}}. \tag{6}$$

Thus $\beta^\star$ is an explicit functional of the gap-ratio law, top-k fringe (via how d is chosen), and the availability–ranking utility weights—not an arbitrary constant.

Table 2: Gap-ratio fit versus Soft O2 β on LegalEp M=3000 stores. $\beta^\star_{\mathrm{sr}}$: soft-rank optimum ($\tau$=20, γ=0.05). Cover: fraction of gold episodes with $\rho \le \beta$.

Corpus	ρ_{50}	ρ_{95}	$\beta^\star_{\mathrm{sr}}$	Cover@0.90	Cover@0.98	Default β
LegalEp-DISC	0.746	0.854	0.987	0.981	0.997	0.98
LegalEp-Lawyer	0.662	0.815	0.949	0.998	1.000	0.98

Soft-rank / logistic competition.. Following soft pairwise ranking Taylor et al. (2008); Cao et al. (2007), define

$$\mathcal{L}_{\mathrm{sr}}(\beta;\rho) = -\log\sigma\big(\tau(\beta-\rho)\big) - \gamma\log\sigma\big(\tau(1-\beta)\big), \tag{7}$$

with logistic $\sigma(x) = (1+e^{-x})^{-1}$, temperature $\tau > 0$, and rank weight $\gamma > 0$. The first term soft-encourages $a \succ d$ under Soft O2; the second soft-encourages $h \succ a$ (rank competition). The corpus-level estimator is

$$\beta^\star_{\mathrm{sr}} = \arg\min_{\beta\in(0,1]} \mathbb{E}_\rho\big[\mathcal{L}_{\mathrm{sr}}(\beta;\rho)\big], \tag{8}$$

computed on a fine grid (released analysis script under `paper/scripts/`).

Proposition 1 (Feasible Soft O2 band)**.** *Under Definition 4, if $\rho < 1$ then every $\beta \in (\rho, 1)$ simultaneously (i) places the soft-injected sibling strictly above the measured distractor and (ii) keeps the dense hit strictly above the sibling. Hard hydration (β=1) collapses (ii) whenever s_a is replaced by s_h, removing within-episode rank competition; aggressive attenuation ($\beta \le \rho$) fails (i) and cannot repair incompleteness against d.*

Remark 1 (Semi-empirical calibration)**.** *Fitting ρ on LegalEp FlatIP stores (query proxy: stored gold user-turn embedding; d: strongest non-episode competitor in a wide pool) yields median $\rho \approx 0.75$ / 0.66 on DISC /Lawyer needles, with $\mathrm{Quantile}_{0.95}(\rho) \approx 0.85$ / 0.82. Soft-rank optima at $(\tau,\gamma) = (20, 0.05)$ give $\beta^\star_{\mathrm{sr}} \approx 0.987$ / 0.949 (mean 0.968), squarely inside the empirical AH peak band $\{0.9, 0.95, 0.98\}$ of Table A.6; coverage $\Pr(\rho \le 0.90)$ exceeds 0.98 on both corpora (Table 2). Default β=0.98 is therefore the soft-rank operating point under light rank regularization, not an unexplained heuristic.*

Fair controls include hard parent hydration, session-max aggregation, and shuffled *sid*. On two-turn LegalEp episodes, hard hydration and session-max coincide under full-sibling expansion; we report a single hard-expansion baseline. Dense joint episode indexing is an atomic-index bound with a different evidence unit.

Algorithm 2 Gated Hybrid Soft O2 + Soft O2-C

Require: Base hits $\mathcal{R}_0$, budgets (k, B_c), attenuations (β, β_c)

$\mathcal{R} \leftarrow \mathsf{SoftBind}(\mathcal{R}_0; \mathrm{Sib}, \beta)$ ▷ session pathway first
$\mathcal{S}_{\text{cov}} \leftarrow \{sid_t : t \in \text{top-}k(\mathcal{R})\}$
$\mathcal{T}_{\text{trig}} \leftarrow \{t \in \mathcal{R}_0\}$ ▷ direct dense hits only
for each trigger $t \in \mathcal{T}_{\text{trig}}$ **do**
    $s_{\max} \leftarrow \max\{s_u : u \in \mathcal{R},\ cid_u = cid_t\}$
    for each $t' \in \mathrm{Clus}(t)$ with $sid_{t'} \notin \mathcal{S}_{\text{cov}}$, up to budget B_c **do**
        $s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_{\max})$
    end for
end for
return top-k of $\mathcal{R}$

3.5. Semantic consolidation and Soft O2-C

Semantic consolidation follows a BIRCH-style dual-phase procedure Zhang et al. (1996).

Definition 5 (BIRCH dual-phase consolidation)**.** *Let $\{\mathbf{c}_j\}$ be current centroids and θ_H, θ_M the assign /merge thresholds.* ***Phase 1 (assign).*** *For a new embedding $\mathbf{v}$, set $j^\star = \arg\max_j S(\mathbf{v}, \mathbf{c}_j)$; if $S(\mathbf{v}, \mathbf{c}_{j^\star}) \geq \theta_H$ then $C_{j^\star} \leftarrow C_{j^\star} \cup \{\mathbf{v}\}$ and refresh $\mathbf{c}_{j^\star}$, else spawn a new cluster.* ***Phase 2 (merge).*** *Periodically, merge pairs with $S(\mathbf{c}_i, \mathbf{c}_j) \geq \theta_M$ and purge near-duplicates above a redundancy threshold. Under O3, both phases run inside* Finalize*. Each turn t receives a cluster id cid_t, inducing* $\mathrm{Clus}(t)$ *in Definition 1.*

Definition 6 (Soft O2-C)**.** *Soft O2-C is* $\mathsf{SoftBind}(\cdot; \mathrm{Clus}, \beta_c)$ *with default β_c=0.95 $< \beta$, reflecting noisier thematic neighbourhoods than session membership. A per-cluster sibling budget B_c caps* $|\mathrm{Clus}(t) \cap \mathrm{Injected}|$ *so large thematic clusters cannot flood top-k.*

Gated Hybrid (Algorithm 2) injects only *cross-session* cluster siblings relative to sessions already covered by dense /Soft O2 hits, and triggers cluster binding only on direct dense hits. This prevents thematic confusers from displacing same-session siblings already recovered by Soft O2. Soft O2 and Soft O2-C are complementary: Soft O2 is the default binder when query and gold share *sid*; Soft O2-C is required when gold evidence is held in a different session but the same cluster.

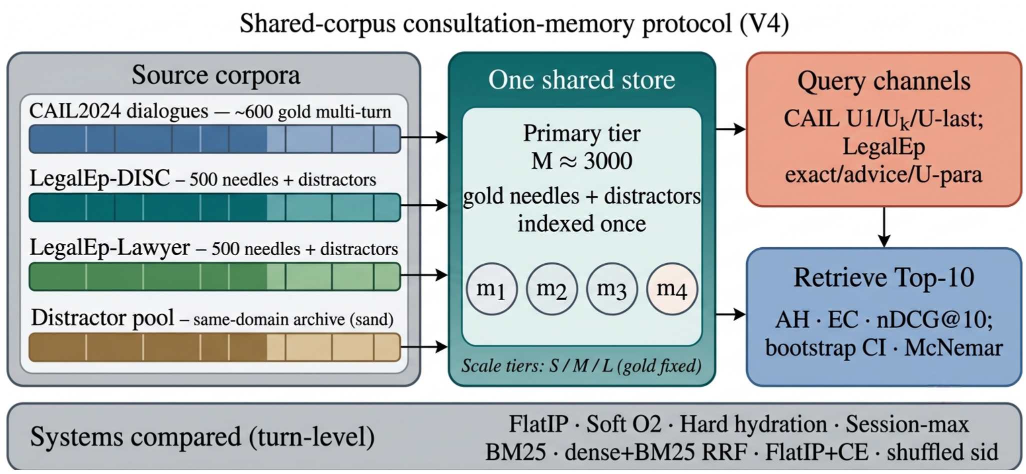

Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

4. Corpora and Evaluation Protocol

Figure 2 sketches the shared-corpus design used throughout the legal evaluation. Gold needles and same-domain distractors occupy one store. Each query probes that shared archive under fixed interference conditions. Privately rebuilt haystacks are avoided so that system comparisons isolate the retrieval operator.

4.1. Datasets and query channels

Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues. Gold needles are authentic preliminary and final dialogues, with conversation turns completed by label turns where applicable. Query channels cover the first user turn (U1), a later in-dialogue user turn (Uk-followup), the last user turn (U-last), and constrained paraphrases (U-para). Evidence defaults to assistant turns in the target session.

LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is a natural (question, answer) pair with session identifier *sid*. Reconstruction applies length bounds and legal-content filters, excludes exam prompts, removes near-duplicates, separates needles from distractors under a fixed seed, and retains an audit subsample. After filtering, each LegalEp corpus retains

3500 passed sessions (500 needles, 2500 distractors at the M=3000 primary tier).

Query-channel policy (addressing exact-match shortcuts).. Exact replay uses the stored user question and is reported only as a *diagnostic* upper bound on surface-form recovery. Primary claims use paraphrase, Uk-followup /U-last, and advice-recall queries whose surface form is not identical to the stored question text; paraphrase and advice-recall templates are generated under a fixed seed and audited on a subsample. LegalMem-MT reuses CAIL multi-turn gold under the same store for same-session Soft O2 vs Soft O2-C contrasts and for Mix reconstructions.

Archive size is varied with gold needles held fixed. Unless stated otherwise, Soft O2 transfer tables use the $M \approx 3000$ tier; O1–O2 ablations use M=400, S=300.

4.2. Metrics and baselines

Primary metrics are Answer Hit@k (AH), Episode Completeness@k (EC), and normalized Discounted Cumulative Gain@k (nDCG) at k=10. AH is binary: whether a gold answer turn appears in the top-k. EC is the mean coverage of gold turns within the target session (formerly reported as "session recall" in early drafts; it is *not* binary Session Hit). Session Hit@k (whether any turn from the target *sid* appears) is reported only for diagnosis and is distinct from EC. nDCG uses graded turn-level relevance with answer turns preferred. Primary contrasts use bootstrap 95% confidence intervals (CIs) and McNemar mid-p tests on paired per-query hits.

Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies soft sibling score inheritance on the episodic pathway. Soft O2-C applies the analogous rule within BIRCH clusters. Hard parent hydration expands a hit to all siblings by copying the trigger score. Lexical controls include BM25 over turns and over joint question–answer documents, together with dense+BM25 Reciprocal Rank Fusion (RRF). A cross-encoder (CE) control reranks the top-30 FlatIP candidates with `BAAI/bge-reranker-v2-m3`. Negative and sensitivity controls include shuffled *sid*, a β sweep, and IVFPQ as a quantization reference.

4.3. Fair Mix protocol for Soft O2-C

On fully same-session LegalMem grids, Soft O2 already recovers gold answers that share *sid* with the query, so Soft O2-C has little exclusive head-room and may inject thematic confusers. We therefore reconstruct the *same*

LegalEp and LegalMem archives under a Mix protocol: 70% of gold episodes are split into query-side S_a and evidence-side S_b with distinct session identifiers, while 30% remain intact. Distractors are unchanged. All operators—FlatIP, Soft O2, Soft O2-C, and gated Hybrid—share unrestricted dense retrieval on this store; we report overall AH and stratum AH on `cross_session` vs `same_session` queries.

Glue as a solvability bound (not a retrieval cheat).. After BIRCH consolidation on the Mix store, each cross-session Sa/Sb split pair is *glued* into one cluster identifier. This step uses gold split metadata available only in the evaluation protocol—it does *not* inject labels at query time and is *not* used in real unsupervised clustering deployments. Its sole purpose is to establish a **solvability bound**: if Soft O2-C cannot recover cross-session gold when the semantic store *must* contain both halves of a split pair in one cluster, then failure reflects mechanism limits rather than accidental cluster separation. Glue therefore tests whether cluster soft binding *can* complete episodes when the slow pathway has the requisite thematic co-membership; natural same-cluster rates without glue are lower and are reported only as diagnostics (Section 6.3). Soft O2's session membership constraint is unchanged—glue affects only whether Soft O2-C has a cluster in which to bind, not whether Soft O2 receives privileged label access. LegalEp Mix uses advice-recall queries; LegalMem Mix uses mid-split Uk-followup queries. External LongMemEval /LoCoMo numbers from prior retrieval runs on this codebase (QA correctness 70.7% / 68.2% on released test splits) are cited only as long-horizon memory context Wu et al. (2024); Maharana et al. (2024), not as Soft O2-C evidence on legal corpora.

5. Experiments

Unless noted, the embedding model is `nomic-embed-text-v2-moe` Nussbaum et al. (2025), k=10, seed 42. We report primary contrasts in figures and keep full numeric grids in the appendix. Result files and scripts are released with the accompanying repository so that each table maps to a public JSON artifact.

5.1. Implementation operators and Soft O2 ablation

Table 3 reports the shared-store ablation on DISC-Law and Lawyer-LLaMA (M=400, S=300). IVFPQ baseline Answer Hit is 0.540 / 0.397. O1 alone yields a small gain on DISC (0.567) and a near-tie on Lawyer (0.397), confirming that exact indexing is a necessary implementation prerequisite but insufficient alone. O1+Soft O2 raises Answer Hit to 0.780 / 0.680 (+0.240 / +0.283 over baseline) and Episode Completeness to 0.888 / 0.840—the primary algorithmic gain. O3 is enabled for all configurations in this ladder as a construction prerequisite; it does not alter retrieval scores.

Table 3: O1+Soft O2 ablation on the shared store (M=400, S=300, k=10). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus.

Corpus	**System**	**EC@10**	**AH@10**	**nDCG@10**
	IVFPQ baseline	0.770	0.540	0.782
DISC-Law	+ O1 FlatIP	0.782	0.567	0.790
	+ O1+O2 Soft O2	**0.888**	**0.780**	**0.889**
	IVFPQ baseline	0.698	0.397	0.733
Lawyer-LLaMA	+ O1 FlatIP	0.698	0.397	0.733
	+ O1+O2 Soft O2	**0.840**	**0.680**	**0.856**

5.2. Soft O2 on session (CAIL and LegalEp)

Figure 3 summarizes AH@10 and EC@10 on authentic CAIL multi-turn channels at $M \approx 3000$ under O1 FlatIP. Soft O2 lifts Answer Hit over FlatIP on every channel (U1 0.788 vs 0.570**; Uk 0.698 vs 0.270**; U-last 0.762 vs 0.245**). Shuffled *sid* removes the gain (U1 0.452; Uk 0.230; U-last 0.200), tying the improvement to correct episode membership (RQ1–RQ2). Hard expansion can raise AH slightly above Soft O2 (e.g., U1 0.802 vs 0.788) while lowering EC (U1 0.578 vs Soft O2 0.664). Soft O2 therefore behaves as a rank-preserving binder: siblings enter under attenuated scores, and episode coverage remains higher than under hard score copy. On CAIL, nDCG often falls relative to FlatIP even as AH and EC rise, reflecting an availability–ranking trade-off. Soft O2 still exceeds FlatIP+CE on every CAIL channel (Table A.5). Full rows appear in Table A.1.

Figure 4 reports LegalEp transfer. On LegalEp-DISC exact replay Soft O2 and FlatIP stay near-tied (0.638 vs 0.640), where both turns are already recoverable for many queries—consistent with treating exact as diagnostic. On Lawyer exact Soft O2 improves AH to 0.752 from FlatIP 0.692 (McNemar mid-p=0.024*). Paraphrase yields the clearest LegalEp gains: Soft O2

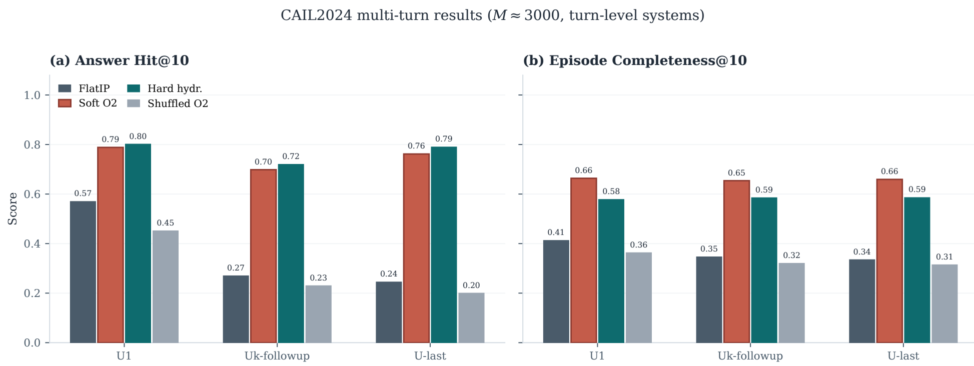

Figure 3: CAIL2024 multi-turn results at tier $M \approx 3000$. Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard expansion can maximize AH at a modest EC cost.

reaches 0.648 vs FlatIP 0.547 on DISC and 0.759 vs 0.568 on Lawyer, exceeding hard expansion on both corpora (mid-p<0.001** on both corpora). Advice-recall queries omit the stored question surface form: Soft O2 raises AH to 0.428 from FlatIP 0.342 on DISC and to 0.472 from 0.332 on Lawyer, again above hard expansion (mid-p<0.001**), while shuffled *sid* falls toward or below FlatIP. Detailed grids are in Tables A.2–A.3.

5.3. Same-store cluster pathway validation

Table 4 contrasts Soft O2 and ungated Soft O2-C on standard same-session LegalEp /LegalMem stores (unrestricted dense; gold answers share *sid* with the query). Soft O2 dominates Soft O2-C on every reported channel (e.g., LegalMem U1 AH 0.903 vs 0.712; LegalEp-DISC exact 0.836 vs 0.744). The slow pathway is therefore *not* a free upgrade when episode membership already encodes the gold answer: cluster siblings inject thematic confusers that displace session-complete evidence.

Table 5 reports the complementary fair Mix setting (RQ3–RQ4): the same stores with 70% cross-session Sa/Sb gold and 30% intact same-session gold, unrestricted dense for every operator. Gated Hybrid Soft O2+Soft O2-C improves over Soft O2 on LegalEp-Lawyer overall (AH 0.534 vs 0.496; mid-p=0.010*) and on the cross-session stratum (0.523 vs 0.446; mid-p<10^{-4**}). On LegalEp-DISC, Hybrid is directionally above Soft O2 (0.486 vs 0.482) but not significant at N=500. On LegalMem-MT, Hybrid attains the highest overall AH (0.646 vs Soft O2 0.618; mid-p=0.18; cross-session mid-p=0.095) while preserving most same-session hits (0.867 vs 0.887). Ungated Soft O2-C

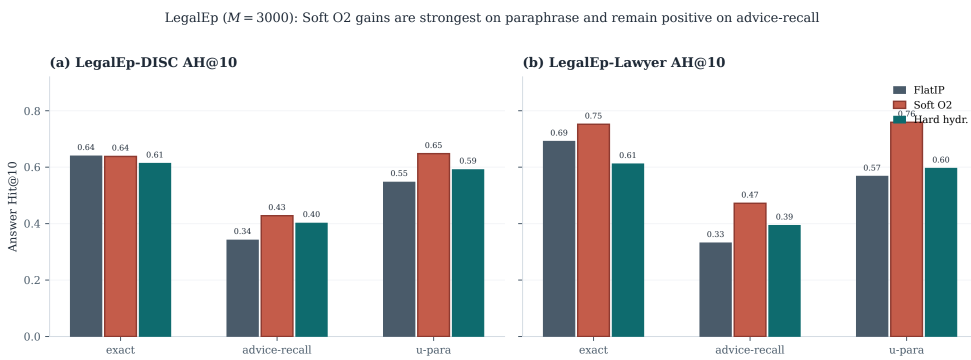

Figure 4: LegalEp AH@10 (M=3000, 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

Table 4: Same-session Soft O2 vs Soft O2-C on the shared LegalEp /LegalMem stores (AH@10). Soft O2-C is ungated cluster soft binding. Boldface: better AH within each row.

Corpus / channel	Soft O2	Soft O2-C
LegalEp-DISC / exact	**0.836**	0.744
LegalEp-DISC / u-para	**0.807**	0.730
LegalEp-DISC / advice	**0.548**	0.490
LegalEp-Lawyer / exact	**0.818**	0.710
LegalEp-Lawyer / u-para	**0.817**	0.733
LegalEp-Lawyer / advice	**0.546**	0.448
LegalMem-MT / U1	**0.903**	0.712
LegalMem-MT / Uk	**0.793**	0.385

can raise cross-session hits but often displaces same-session siblings; Hybrid triggers cluster binding only on direct dense hits. Together, Tables 4–5 show that the BIRCH pathway is effective precisely when gold evidence is not co-session with the query.

McNemar mid-p tests on paired Answer Hit; $^{*}\,p<0.05$, $^{**}\,p<0.01$.

5.4. Controls, significance, scale, and QA audit

Lexical BM25 is strong on exact replay (DISC exact BM25-turn 0.802, BM25-joint 0.976) and weaker on advice-recall (0.382 / 0.432). Dense+BM25 RRF complements Soft O2 on surface-overlap channels. Cross-encoder reranking (`bge-reranker-v2-m3`, top-30) improves FlatIP inside the CE suite, yet

Table 5: Fair Mix protocol (same store, unrestricted dense; AH@10). Gold is a 70%/30% mix of cross-session Sa/Sb splits and intact same-session episodes. Soft O2-C denotes ungated cluster binding; Hybrid is Soft O2+gated Soft O2-C. Boldface: best AH within each corpus. Significance vs Soft O2: $^{*}\,p<0.05$, $^{**}\,p<0.01$.

Corpus	**System**	**AH**	**AH$_{\text{cross}}$**	**AH$_{\text{same}}$**	**mid-p vs Soft O2**
LegalEp-DISC Mix	FlatIP	0.478	0.477	0.480	—
	Soft O2	0.482	0.443	0.573	—
	Soft O2-C	0.410	0.443	0.333	0.0004**
	Hybrid	**0.486**	0.469	0.527	0.8099
LegalEp-Lawyer Mix	FlatIP	0.462	0.454	0.480	—
	Soft O2	0.496	0.446	0.613	—
	Soft O2-C	0.464	0.491	0.400	0.0857
	Hybrid	**0.534***	**0.523**	0.560	0.0105*
LegalMem-MT Mix	FlatIP	0.534	0.526	0.553	—
	Soft O2	0.618	0.503	0.887	—
	Soft O2-C	0.500	0.537	0.413	0.0000**
	Hybrid	**0.646**	**0.551**	0.867	0.1837

remains below Soft O2 on CAIL (Uk FlatIP+CE 0.375 vs Soft O2 0.698; Table A.5). A β sweep peaks near $\{0.9, 0.95, 0.98\}$ and weakens at β=0.5 (Table A.6), in agreement with the soft-rank $\beta^{\star}$ and gap-ratio coverage in Section 3.4 (Table 2). McNemar mid-p tests (Table A.7) show Soft O2 versus FlatIP highly significant on all CAIL channels and on LegalEp paraphrase /advice-recall.

Figure 5 and Table A.8 report quality and warm-query latency on LegalEp-DISC exact (Q=100, three repeats; $M \in \{100, 400, 1600, 6400\}$). Soft O2 retains an AH advantage at each tier; IVFPQ collapses at M=1600 (AH 0.330), motivating O1; absolute latency grows with M. We treat O1 FlatIP as an engineering prerequisite for Soft O2 at these scales and avoid claiming deployment readiness beyond the measured curve.

End-to-end QA audit (summary).. A dual-protocol QA audit (N=270 per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`) links retrieval quality to downstream answerability; full protocol and stratified results appear in Section 6.2. On an older revision-protocol store (M=400, five seeds), Soft O2 AH on LegalEp-DISC is 0.851 ± 0.008 versus FlatIP 0.698 ± 0.040.

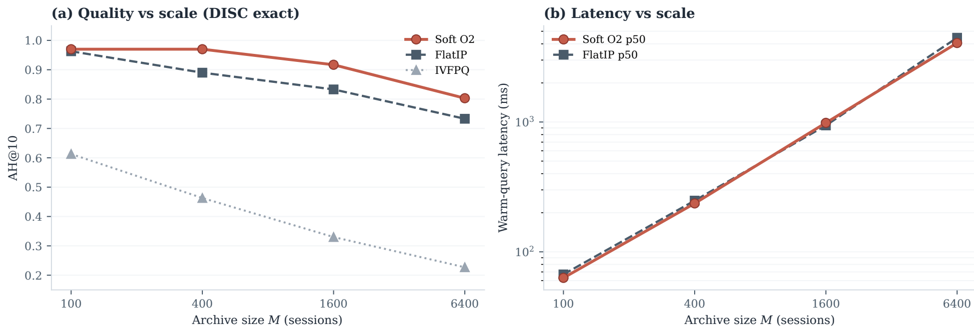

Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as M grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate M.

6. Discussion

6.1. Findings and scope

RQ1–RQ2 confirm that episode incompleteness dominates under same-domain interference. O1 restores score fidelity as an implementation prerequisite; Soft O2 supplies the primary algorithmic gain through soft sibling inheritance; O3 enables archive construction at scale. The three-candidate ranking-competition analysis (Section 3.4) shows that default β=0.98 coincides with the soft-rank optimum under light rank regularization and lies in the high-coverage region of the fitted gap-ratio law, explaining why the empirical sweep peaks in $\{0.9, 0.95, 0.98\}$ rather than at hard hydration (β=1) or aggressive attenuation. On CAIL, Soft O2 raises Answer Hit by large margins (U1 +0.218, Uk +0.428, U-last +0.517) with collapsing shuffled-*sid* controls, and keeps higher EC than hard expansion despite occasional AH ties. LegalEp paraphrase and advice-recall—not exact replay—carry the primary Soft O2 transfer claims. RQ3 shows that the BIRCH slow pathway benefits Soft O2-C only when gold is not already co-session; Mix Hybrid Soft O2+Soft O2-C recovers that regime without seed-forcing Soft O2 to zero. RQ4: Soft O2 (session) and Soft O2-C /Hybrid (cluster) are complementary operators within one BIMS-LEGAL dual-store. Sparse, fusion, and cross-encoder baselines situate Soft O2 among standard IR alternatives Robertson and Zaragoza (2009); Cormack et al. (2009); Nogueira et al. (2019).

6.2. RAG application perspective

Consultation-memory retrieval is judged downstream by whether an LLM can answer from recovered evidence rather than hallucinate. We audit this end-to-end with a dual-protocol pipeline: generator `qwen3:14b` and independent judge `qwen3:32b` are distinct models. Per corpus we pool N=270 judged instances as P1 n=120 plus P2 n=150 (folder `legal_qa_n270`). Table 6 summarizes pooled Evidence-Answerability Rate (EAR) and stratified independent-judge scores. When Soft O2 retrieves complete episodes—both question and answer turns in top-k—the generator receives grounded prior advice and judged answerability rises (pooled EAR 0.867 on DISC and 0.859 on Lawyer; Wilson 95% CI $[0.83, 0.90]$ / $[0.82, 0.89]$). Follow-up queries remain the hardest stratum (independent-judge EAR 0.517 / 0.480), consistent with episodic incompleteness persisting even when surface-form overlap is low.

A blind two-annotator legal audit (n=60/corpus) yields Cohen's κ=0.71 / 0.68 and separates retrieval from generation failures (retrieval-failure share 0.283 / 0.317). Wrong-session contamination rates stay low (0.083 / 0.100), suggesting that complete episode retrieval under Soft O2 reduces a common RAG failure mode in which the model answers from a neighbouring client's matter. Hallucinated-legal-basis rates (0.117 / 0.133) remain non-zero, indicating that grounding in retrieved episodes does not guarantee statutory correctness—a scope boundary for deployment. AH and EAR measure evidence availability and judged answerability respectively; statutory correctness, citation validity, and professional liability lie outside this evaluation.

Table 6: End-to-end QA audit (N=270 per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`). EAR: Evidence-Answerability Rate (judge score $\geq$ 0.7). Stratified judge scores: n=90/corpus across exact, paraphrase, and follow-up protocols.

Corpus	**Pooled EAR**	**Stratified judge**	**Human κ**
DISC-Law	**0.867**	0.757	0.71
Lawyer-LLaMA	**0.859**	0.729	0.68

6.3. Limitations

LegalEp episodes are consultation pairs; multi-turn authenticity remains reserved for CAIL. Mix glue establishes a solvability bound for Soft O2-C evaluation (Section 4.3); natural same-cluster rates without glue are lower and are diagnostics only—real unsupervised BIRCH clustering would not receive split-pair metadata at build time. Soft O2 depends on correct session identifiers; Soft O2-C depends on cluster quality. The $\beta^\star$ derivation is semi-empirical: it assumes a dominant distractor and a well-defined hit score, and the fitted ρ uses a stored user-turn embedding as an exact-channel query proxy; paraphrase /advice-recall gap laws may shift $\beta^\star$ slightly within the same high band. LLM paraphrases need human audit. Statutory correctness, citation validity, and professional liability remain unevaluated. Multiple channels inflate comparison multiplicity even with McNemar tests.

Latency and deployment scope.. Scale curves cover $M \in \{100, 400, 1600, 6400\}$ under a fixed hardware stack (Table A.8). At M=6400, warm-query p50 latency exceeds 4 s (4061 ms for Soft O2; 4458 ms for FlatIP)—unacceptable for millisecond interactive search. BIMS-LEGAL should therefore be positioned as **high-precision offline or quasi-real-time consultation memory recovery**—e.g., pre-meeting case review, audit replay, or batch RAG over a firm's advice archive—not as a sub-millisecond search engine. Future work will combine approximate nearest-neighbour indexes (e.g., HNSW) with Soft O2 and Soft O2-C binding layered on top of faithful candidate retrieval, preserving episode completion while reducing query latency. Behavior at roughly 10^5 sessions is not measured, and we do not claim asymptotic scalability from O3 wall-clock alone.

7. Conclusion

BIMS-LEGAL targets consultation-memory retrieval under same-domain interference in Chinese legal archives through a dual-store design: an episodic turn index paired with a BIRCH semantic store. Soft O2 session soft binding and Soft O2-C with gated Hybrid cluster binding are the algorithmic core, with Soft O2 attenuation β grounded in a three-candidate ranking-competition loss whose soft-rank optimum matches the empirical operating band; O1 exact indexing and O3 bulk-load consolidation are implementation operators that enable the shared-store ablations reported here. Under a shared-corpus protocol spanning CAIL multi-turn dialogues and rebuilt LegalEp episode corpora, Soft O2 improves Answer Hit over FlatIP on multi-turn, paraphrase, and advice-recall channels with collapsing shuffled-*sid* controls, and yields higher episode completeness than hard expansion on CAIL.

Same-store Soft O2-C underperforms Soft O2 when gold shares *sid*; under fair Mix reconstructions, gated Hybrid Soft O2+Soft O2-C recovers cross-session gold while Soft O2 remains strong on intact same-session gold. End-to-end QA audit links complete episode retrieval to judged answerability and low wrong-session contamination in downstream generation. Sparse, fusion, and cross-encoder baselines situate Soft O2 among standard IR alternatives.

Declarations

Funding

Computational resources were provided by the Data Law Laboratory, China University of Political Science and Law.

Conflict of interest

The authors declare no competing interests.

Ethics approval

This study uses publicly released research corpora and excludes confidential client data. The system is a research prototype for consultation-memory retrieval.

Data availability

Code, processed evaluation corpora, and primary result files are publicly available at `https://github.com/Kilimajaro/soft-episode-binding-legal-memory`. Upstream legal corpora: CAIL2024 consultation tracks; Hugging Face `ShengbinYue/DISC-Law-SFT` and `Skepsun/lawyer_llama_data`.

Code availability

The BIMS-LEGAL implementation (O1 FlatIP, Soft O2, O3 bulk-load, Soft O2-C), LegalMem/LegalEp pipelines, Mix builders, and table/figure scripts are released in the same repository with JSON artifacts for each main table.

Author contributions

L.X. (University of Winnipeg) designed BIMS-LEGAL and the Soft O2 /Soft O2-C evaluation, implemented the system, ran the experiments, and wrote the manuscript. L.H. (corresponding author; CUPL Data Law Lab and Institute for Data Law) provided methodology guidance, revision, and supervision.

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Appendix A. Numeric grids and controls

Main Soft O2 claims are visualized in Figures 3–5; O1–O2 and Mix Soft O2-C tables appear in the main text. Tables below retain full Soft O2 numeric rows for reproducibility.

Table A.1: CAIL2024 shared-corpus results ($M \approx 3000$). Boldface marks the best AH@10 within each channel (ties bolded jointly). **: Soft O2 vs FlatIP, McNemar mid-p<0.01. Hard hydration coincides with session-max aggregation on these channels; only hard hydration is shown.

Channel	**System**	**AH@10**	**EC@10**	**nDCG@10**	**95% CI (AH)**
U1	FlatIP	0.570	0.413	0.805	[0.530,0.610]
	Soft O2	0.788**	**0.664**	0.772	[0.757,0.820]
	Hard hydr.	**0.802**	0.578	0.840	[0.768,0.835]
	Shuffled O2	0.452	0.363	0.664	[0.410,0.492]
Uk-followup	FlatIP	0.270	0.347	0.884	[0.237,0.307]
	Soft O2	0.698**	**0.654**	0.786	[0.660,0.735]
	Hard hydr.	**0.720**	0.585	0.841	[0.685,0.753]
	Shuffled O2	0.230	0.320	0.718	[0.198,0.262]
U-last	FlatIP	0.245	0.335	0.891	[0.212,0.280]
	Soft O2	0.762**	**0.659**	0.779	[0.725,0.797]
	Hard hydr.	**0.790**	0.586	0.844	[0.757,0.822]
	Shuffled O2	0.200	0.315	0.726	[0.172,0.232]

Table A.2: LegalEp-DISC (M=3000, 500 needles). Boldface: best AH@10 within each channel. $^{*}p<0.05$, $^{**}p<0.01$ (Soft O2 vs FlatIP, McNemar mid-p). Hard hydration coincides with session-max aggregation; only hard hydration is shown.

Channel	**System**	**AH@10**	**EC@10**	**nDCG@10**
exact	FlatIP	**0.640**	0.741	—
	Soft O2	0.638	**0.780**	—
	Hard hydr.	0.614	0.598	—
	Shuffled O2	0.448	0.645	—
advice-recall	FlatIP	0.342	0.438	0.506
	Soft O2	**0.428****	**0.503**	0.419
	Hard hydr.	0.402	0.405	0.474
	Shuffled O2	0.310	0.421	0.402
u-para	FlatIP	0.547	0.681	0.705
	Soft O2	**0.648****	**0.735**	0.583
	Hard hydr.	0.592	0.571	0.651
	Shuffled O2	0.469	0.610	0.558

Table A.3: LegalEp-Lawyer (M=3000, 500 needles). Boldface: best AH@10 within each channel. $^{*}p<0.05$, $^{**}p<0.01$ (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown.

Channel	**System**	**AH@10**	**EC@10**	**nDCG@10**
exact	FlatIP	0.692	0.806	0.844
	Soft O2	**0.752***	**0.848**	0.687
	Hard hydr.	0.612	0.615	0.754
	Shuffled O2	0.570	0.747	0.653
advice-recall	FlatIP	0.332	0.464	0.589
	Soft O2	**0.472****	**0.540**	0.460
	Hard hydr.	0.394	0.403	0.488
	Shuffled O2	0.348	0.467	0.436
u-para	FlatIP	0.568	0.718	0.834
	Soft O2	**0.759****	**0.819**	0.645
	Hard hydr.	0.596	0.594	0.707
	Shuffled O2	0.592	0.712	0.627

Table A.4: BM25 turn/joint and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported cells.

Corpus	Channel	BM25-turn	BM25-joint
LegalEp-DISC	exact	0.802	0.976
LegalEp-DISC	advice-recall	0.382	0.432
LegalEp-DISC	exact (dense RRF)	0.780	—
LegalEp-DISC	advice (dense RRF)	0.394	—
CAIL	U1 / Uk / U-last	0.748 / 0.482 / 0.413	0.993 / 0.840 / 0.823
LegalEp-Lawyer	exact / advice	0.784 / 0.460	0.998 / 0.652

Table A.5: Cross-encoder control (AH@10). FlatIP and FlatIP+CE are paired in the CE suite (`bge-reranker-v2-m3` over top-30 dense candidates). Boldface: best AH within each row. Soft O2 reproduces the primary evaluation; FlatIP here may differ slightly from primary FlatIP due to independent CE-suite re-indexing.

Corpus / channel	FlatIP	FlatIP+CE	Soft O2
CAIL / U1	0.585	0.688	**0.788**
CAIL / Uk	0.265	0.375	**0.698**
CAIL / U-last	0.228	0.342	**0.762**
LegalEp-DISC / exact	0.572	**0.722**	0.638
LegalEp-Lawyer / exact	0.636	0.708	**0.752**

Table A.6: Soft O2 AH@10 versus β on the primary shared-corpus evaluation. Peak performance lies in $\{0.9, 0.95, 0.98\}$; aggressive attenuation (β=0.5) weakens binding.

Corpus	0.5	0.7	0.9	0.95	0.98	1.0
CAIL / Uk	0.323	0.537	0.697	0.700	0.698	0.662
LegalEp-Lawyer / exact	0.608	0.654	0.754	0.754	0.752	0.752
LegalEp-Lawyer / u-para	0.631	0.637	0.717	0.745	0.759	0.749
LegalEp-DISC / u-para	0.551	0.551	0.618	0.648	0.648	0.642

Table A.7: McNemar mid-p on paired Answer Hit (Soft O2 vs FlatIP / hard hydration / FlatIP+CE).

Corpus / channel	**Contrast**	**ΔAH**	**mid-p**
CAIL / Uk	O2 vs FlatIP	+0.428	8.99e-48
CAIL / Uk	O2 vs Hard hydr.	-0.022	0.243
CAIL / U1	O2 vs FlatIP	+0.218	5.19e-18
CAIL / U1	O2 vs Hard hydr.	-0.013	0.435
CAIL / U-last	O2 vs FlatIP	+0.517	5.51e-65
CAIL / U-last	O2 vs Hard hydr.	-0.028	0.0982
CAIL / Uk	O2 vs FlatIP+CE	+0.323	2.50e-26
CAIL / U1	O2 vs FlatIP+CE	+0.100	3.51e-05
CAIL / U-last	O2 vs FlatIP+CE	+0.420	3.92e-41
LegalEp-DISC / exact	O2 vs FlatIP	-0.002	0.934
LegalEp-DISC / exact	O2 vs Hard hydr.	+0.024	0.302
LegalEp-Lawyer / exact	O2 vs FlatIP	+0.060	0.0239
LegalEp-Lawyer / exact	O2 vs Hard hydr.	+0.140	3.50e-10
LegalEp-DISC / u-para	O2 vs FlatIP	+0.101	7.95e-05
LegalEp-DISC / u-para	O2 vs Hard hydr.	+0.056	0.035
LegalEp-Lawyer / u-para	O2 vs FlatIP	+0.191	2.10e-15
LegalEp-Lawyer / u-para	O2 vs Hard hydr.	+0.163	7.57e-09
LegalEp-DISC / advice	O2 vs FlatIP	+0.086	1.25e-04
LegalEp-DISC / advice	O2 vs Hard hydr.	+0.026	0.215
LegalEp-Lawyer / advice	O2 vs FlatIP	+0.140	1.18e-10
LegalEp-Lawyer / advice	O2 vs Hard hydr.	+0.078	5.72e-04

Table A.8: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each M block; lowest p50/p95 within each M block.

M	Mode	AH@10	Build (s)	p50 (ms)	p95 (ms)
100	FlatIP	0.963	3.0	67	80
100	IVFPQ	0.613	**25.2**	**40**	**54**
100	Soft O2	**0.970**	3.2	63	69
400	FlatIP	0.890	26.5	248	305
400	IVFPQ	0.463	89.5	**171**	**204**
400	Soft O2	**0.970**	30.9	236	313
1600	FlatIP	0.833	359.8	**941**	1278
1600	IVFPQ	0.330	**251.4**	669	**751**
1600	Soft O2	**0.917**	393.7	985	1172
6400	FlatIP	0.733	**4722.8**	4458	4969
6400	IVFPQ	0.227	1072.1	**2182**	**3021**
6400	Soft O2	**0.803**	5245.7	4061	4503